DLD
14.10.25

Counter

counter is used to count pulses & it can be used as freq. divider.

counter
- up counter [0, 1, 2, 3, ... , N.]
- down counter [N, N-1, N-2, ... , 2, 1, 0]

classification of counter
- synchronous counter → all memory elements (Flip Flops) are connected with the same clock.
- asynchronous counter ⟹ all F.F এর জন্য different clk.

n bit Counter

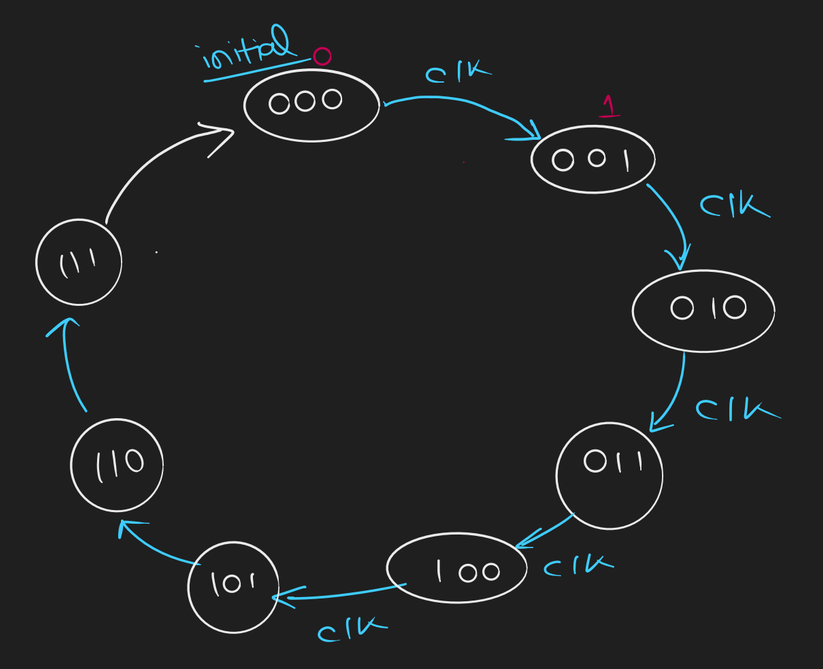
n=1 0, 1 (2টা)
n=2 0, 1, 2, 3 (4টা)
n=3 0, 1, 2, ... , 7. (8টা)

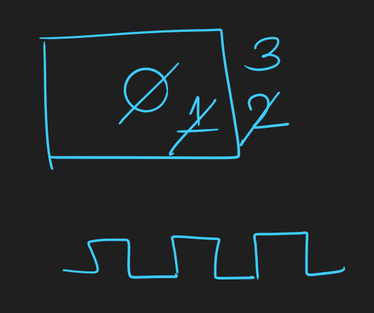
n = N. 0 to $(2^N - 1)$

3 bit counter

state 0 0 0 → 0
state 0 0 1 → 1
0 1 0 → 2
0 1 1 → 3
1 0 0 → 4
1 0 1 → 5
1 1 0 → 6
1 1 1 → 7

up counter ↓
down counter ↑


initial
0
000
clk
1
001
clk
010
clk
011
100
clk
101
clk
110
111



- each clock pulse will make the state change.

3 bit Sync Counter by T Flip Flop

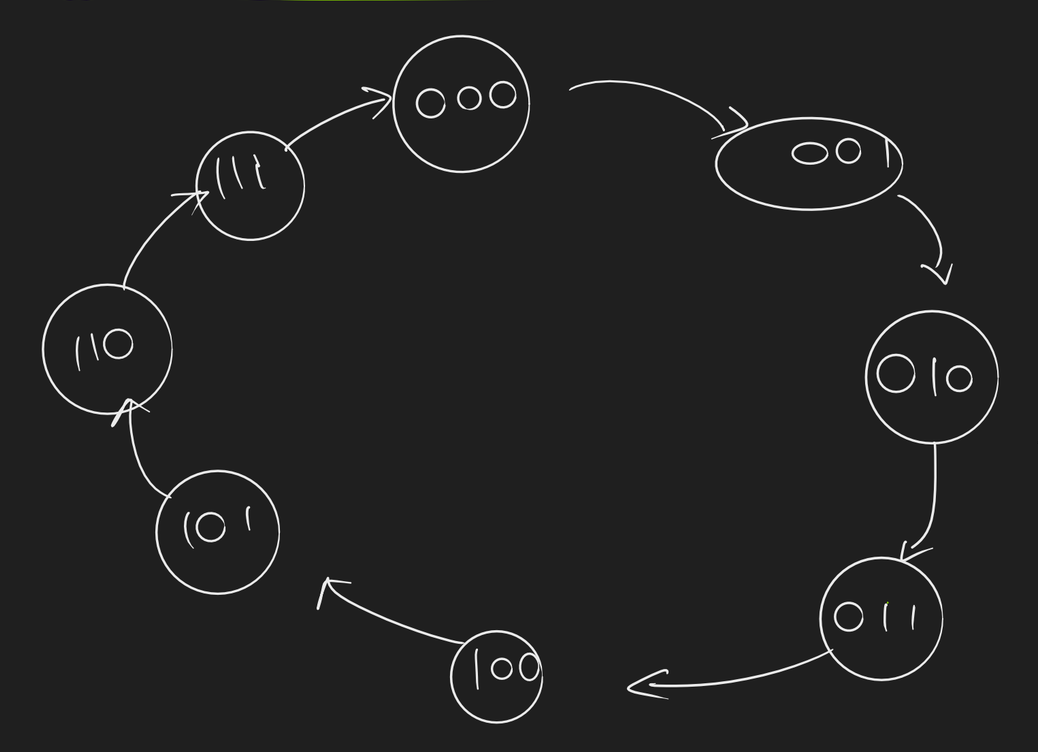
Steps :-
1. Identify number of bits & flip flop.
2. Write excitation table of that ff
   ↳ will be given.
3. Make state diagram & state table.
4. Solve boolean expression (with kMap)
5. Make ckt diagram.

① n=3, FF = T Flip Flop.

(ii)

curr. state Q_n	next state Q_{n+1}	input T
0	0	0
0	1	1
1	0	1
1	1	0

(iii) State diagram


000
001
010
011
100
101
110
111

State Table

curr Q_2	Q_1	Q_0	next Q_2^+	Q_1^+	Q_0°	T_2	T_1	T_0
0	0	0	0	0	1	0	0	1
0	0		0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

4. Find Boolean Expression

T_2

Q_2 \ Q_1Q_0	00	01	11	10
0	^{0}	^{1}	1 ^{3}	^{2}
1	^{4}	^{5}	1 ^{7}	^{6}

$\boxed{T_2 = Q_1 Q_0}$

T_1

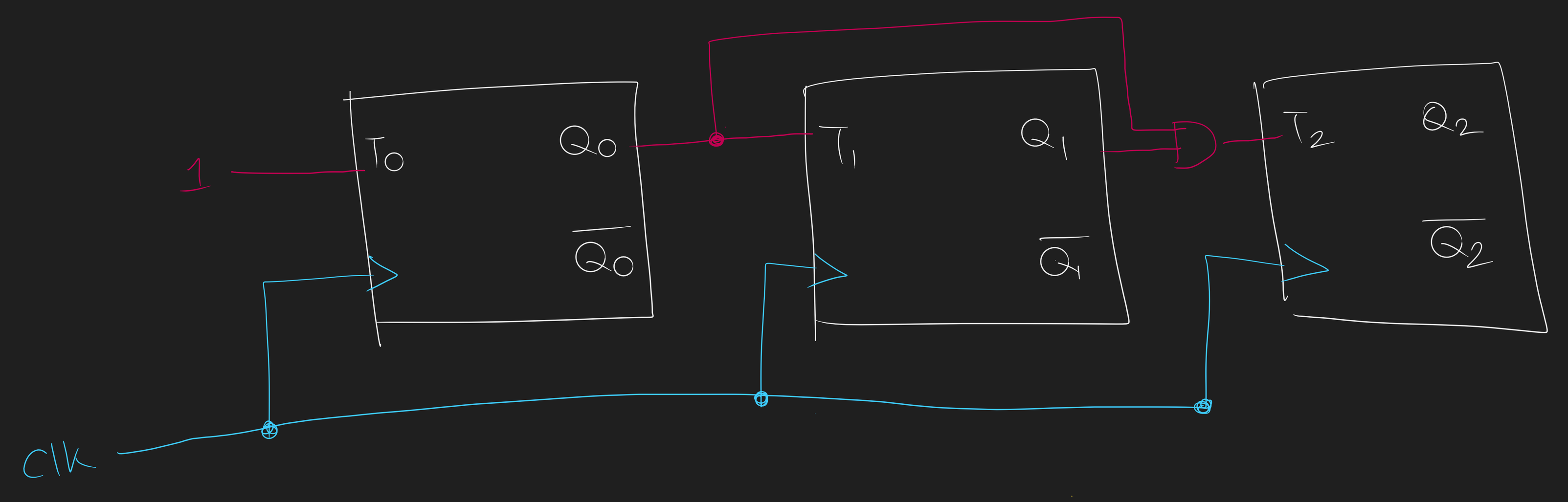
Q_2 \ Q_1Q_0	00	01	11	10
0	^{0}	1 ^{1}	1 ^{3}	^{2}
1	^{4}	1 ^{5}	1 ^{7}	^{6}

$\boxed{T_1 = Q_0}$

T_0

$\boxed{T_0 = 1}$

⑤ ckt diagram


1
T_0
Q_0
$\overline{Q_0}$
T_1
Q_1
$\overline{Q_1}$
T_2
Q_2
$\overline{Q_2}$
Clk